

EXNO:1**DATE :****SHELL PROGRAMMING****Aim:**

To write a shell script to show various system configuration like,

- Currently logged user and his log name
- Your current shell
- Your home directory
- Your operating system type
- Your current path setting
- Your current working directory
- Show Currently logged number of users

Algorithm

1. Start the program.
2. Display the currently logged-in username using whoami.
3. Display the login name using logname.
4. Display the current shell using the \$SHELL variable.
5. Display the home directory using the \$HOME variable.
6. Display the operating system type using uname -o.
7. Display the current PATH setting using the \$PATH variable.
8. Display the current working directory using pwd.
9. Count and display the number of currently logged-in users using who | wc -l.
10. Stop the program.

Procedure**Step 1: Create the Shell Script**

Open the terminal and type: vi systeminfo.sh

Press i to enter Insert Mode.

Type the shell script.

Press Esc to exit Insert Mode.

Save and quit by typing: :wq

Press Enter.

Step 2: Change File Permission

Make the script executable by entering: chmod +x systeminfo.sh

Step 3: Execute the Script

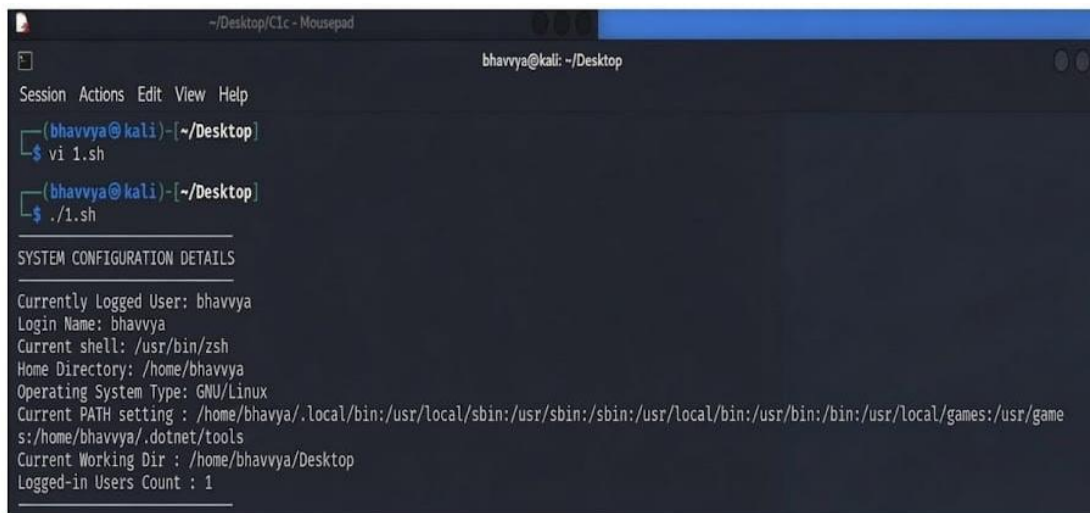
Run the script using: ./systeminfo.sh

Program:

```
#!/bin/bash
echo "-----"
echo "  SYSTEM CONFIGURATION DETAILS"
echo "-----"

echo "Currently Logged User : $(whoami)"
echo "Login Name      : $(logname)"
echo "Current Shell    : $SHELL"
echo "Home Directory   : $HOME"
echo "Operating System Type : $(uname -o)"
echo "Current PATH Setting : $PATH"
echo "Current Working Dir : $(pwd)"
echo "Logged-in Users Count : $(who | wc -l)"

echo "-----"
```

Output:

```
~/Desktop/CLC - Mousepad
bhavvya@kali: ~/Desktop
Session Actions Edit View Help
(bhavvya@kali)-[~/Desktop]
$ vi 1.sh
(bhavvya@kali)-[~/Desktop]
$ ./1.sh
SYSTEM CONFIGURATION DETAILS
Currently Logged User: bhavvya
Login Name: bhavvya
Current shell: /usr/bin/zsh
Home Directory: /home/bhavvya
Operating System Type: GNU/Linux
Current PATH setting : /home/bhavvya/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/home/bhavvya/.dotnet/tools
Current Working Dir : /home/bhavvya/Desktop
Logged-in Users Count : 1
```

Result:

Thus, the shell script is executed and the output is verified.

EXNO:2**DATE :****SHELL PROGRAMMING****Aim:**

Write shell script to show various system configuration like,

- About your OS and version, release number, kernel version
- Show all available shells
- Show mouse settings
- Show computer CPU information like processor type, speed etc
- Show memory information
- Show hard disk information like size of hard-disk, cache memory, model etc
- File system (Mounted)

Algorithm

1. Start the shell script.
2. Display the operating system details, including the OS name, version, release number, and kernel version.
3. Display the list of all available login shells in the system.
4. Display the mouse and other pointing device settings.
5. Retrieve and display CPU information, including processor architecture, model, number of CPUs, and clock speed.
6. Retrieve and display memory information, including total, used, and available memory.
7. Retrieve and display hard disk information, including disk size, model, and partition details.
8. Display the list of all mounted file systems along with their storage usage.
9. End the shell script.

Program

```
#!/bin/bash

echo "===== "
echo "  SYSTEM CONFIGURATION REPORT"
echo "===== "

echo
echo "1. Operating System Information"
cat /etc/os-release

echo
echo "Kernel Version"
uname -r
```

```
echo
echo "===== "

echo "2. Available Shells"
cat /etc/shells

echo
echo "===== "

echo "3. Mouse Settings"
xinput --list

echo
echo "===== "

echo "4. CPU Information"
lscpu

echo
echo "===== "

echo "5. Memory Information"
free -h

echo
echo "===== "

echo "6. Hard Disk Information"
lsblk

echo
echo "===== "

echo "7. Mounted File Systems"
df -h

echo
echo "===== "
```

Output:

```

~/Desktop/c1c - Mousepad
Bhavvy@kali: ~/Desktop
Session Actions Edit View Help
(Bhavvy@kali)-[~/Desktop]
$ vi 2.sh
(Bhavvy@kali)-[~/Desktop]
$ ./2.sh

SYSTEM CONFIGURATION REPORT

1. Operating System Information
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.3"
VERSION="2026.3"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

Kernel Version
7.0.12-kali-amd64

2. Available Shells
# /etc/shells: valid login shells
/bin/sh
/usr/bin/sh
/bin/bash
/usr/bin/bash
/bin/rbash
/usr/bin/rbash
/usr/bin/dash

```

```

/usr/bin/pwsh
/opt/microsoft/powershell/7/pwsh
/usr/bin/screen
/usr/bin/tmux
/bin/zsh
/usr/bin/zsh

```

```

3. Mouse Settings
./2.sh: 17: xinput: not found

```

```

Session Actions Edit View Help

4. CPU Information
Architecture:           x86_64
CPU op-mode(s):         32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 1
On-line CPU(s) list:    0
Vendor ID:              AuthenticAMD
Model name:             AMD Ryzen 3 3250U with Radeon Graphics
CPU family:             23
Model:                  24
Thread(s) per core:     1
Core(s) per socket:     1
Socket(s):              1
Stepping:               1
Microcode version:      0x810B109
BogoMIPS:               5071.56
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 c
                        lflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt rdtscp lm constant
                        _tsc rep_good nopl nonstop_tsc cpuid extd_apicid pni pclmulqdq monitor ss
                        se3 fma cx16 sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand
                        lahf_lm cr8_legacy abm sse4a misalignsse 3dnowprefetch cpb ssbd vmmcall f
                        sgsbase bmi1 avx2 bmi2 rdseed adx clflushopt sha_ni arat

Caches (sum of all):
L1d:                    32 KiB (1 instance)
L1i:                    64 KiB (1 instance)
L2:                     512 KiB (1 instance)
L3:                     4 MiB (1 instance)

NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0

Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
Itlb multihit:          Not affected
L1tf:                   Not affected
Mds:                    Not affected
Meltdown:               Not affected
Mmio stale data:        Not affected
Old microcode:          Not affected
Reg file data sampling:  Not affected
Retbleed:               Mitigation; untrained return thunk; SMT disabled
Spec rstack overflow:   Mitigation; SMT disabled
Spec store bypass:      Not affected

```

```

Session  Actions  Edit  View  Help
Spec store bypass:      Not affected
Spectre v1:             Mitigation; usercopy/swaps barriers and __user pointer sanitization
Spectre v2:             Mitigation; Retpolines; STIBP disabled; RSB filling; PBRSE-eIBRS Not affected; BHI Not affected
Srbds:                 Not affected
Tsa:                   Not affected
Tsx async abort:       Not affected
Vmscape:               Vulnerable

5. Memory Information
      total      used      free      shared  buff/cache   available
Mem:   1.9Gi      849Mi      183Mi       14Mi       1.16i       1.16i
Swap:   1.3Gi         0B       1.3Gi

6. Hard Disk Information
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0   25G  0 disk
├─sda1 8:1    0  23.7G  0 part /
├─sda2 8:2    0    1K  0 part
└─sda5 8:5    0   1.3G  0 part [SWAP]
sr0   11:0   1 1024M  1 rom

7. Mounted File Systems
Filesystem      Size  Used Avail Use% Mounted on
udev            950M   0  950M   0% /dev
tmpfs           197M  992K  196M   1% /run
/dev/sda1        24G   15G   7.2G  68% /
tmpfs           985M   4.0K  985M   1% /dev/shm
none            1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs           985M  136K  985M   1% /tmp
none            1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M  124K  197M   1% /run/user/1000

```

Result:

Thus, the shell script to show various system configuration executed and the output is verified.

```

~/Desktop/Clc - Mousepad
bhavya@kali: ~/Desktop

Session Actions Edit View Help

(bhavya@kali)~/Desktop
$ vi 1.sh

(bhavya@kali)~/Desktop
$ ./1.sh

SYSTEM CONFIGURATION DETAILS

Currently Logged User: bhavya
Login Name: bhavya
Current shell: /usr/bin/zsh
Home Directory: /home/bhavya
Operating System Type: GNU/Linux
Current PATH setting : /home/bhavya/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/home/bhavya/.dotnet/tools
Current Working Dir : /home/bhavya/Desktop
Logged-in Users Count : 1

```

```

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Bhavvy@kali: ~/Desktop

Session Actions Edit View Help

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/bin/rbash
/usr/bin/rbash
/usr/bin/dash

```

```

Session Actions Edit View Help

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Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 48 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 1
On-line CPU(s) list: 0
Vendor ID: AuthenticAMD
Model name: AMD Ryzen 3 3200U with Radeon Graphics
CPU family: 23
Model: 24
Thread(s) per core: 1
Core(s) per socket: 1
Socket(s): 1
Stepping: 1
Microcode version: 0x0108109
SnpMIPS:
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt rdtscp lm constant_tsc rep_good nopl nonstop_tsc cpuid extd_apicid pni pclmulqdq monitor ssse3 fma cx16 sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand lahf_lm cr8_legacy abm sse4a misalignsse 3dnowprefetch cpb ssbd vmcall fsgsbase bmi1 avx2 bmi2 rdseed adx clflushopt sha_ni arat

Caches (sum of all):
L1d: 32 KiB (1 instance)
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L2: 512 KiB (1 instance)
L3: 4 MiB (1 instance)

NUMA:
NUMA node(s): 1
NUMA nodes CPU(s): 0
Vulnerabilities:
Gather data sampling: Not affected
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Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
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Mmio stale data: Not affected
Old microcode: Not affected
Ref file data sampling: Not affected
Retbleed: Mitigation; untrained return thunk; SMT disabled
Spec rstack overflow: Mitigation; SMT disabled
Spec store bypass: Not affected

```

```

Session Actions Edit View Help

Spec store bypass: Not affected
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Retpolines; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected; BHI Not affected
Srbds: Not affected
Tsx: Not affected
Tsx async abort: Not affected
Vmscape: Vulnerable

5. Memory Information
NAME total used free shared buff/cache available
Mem: 1.9Gi 840Mi 183Mi 14Mi 1.1Gi 1.1Gi
Swap: 1.3Gi 0 1.3Gi

6. Hard Disk Information
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 256G 0 disk
├─sda1 8:1 0 23.7G 0 part /
├─sda2 8:2 0 1K 0 part
├─sda5 8:5 0 1.3G 0 part [SWAP]
sr0 11:0 1 1024M 1 rom

7. Mounted File Systems
Filesystem Size Used Avail Use% Mounted on
udev 950M 0 950M 0% /dev
tmpfs 197M 992K 196M 1% /run
/dev/sda1 24G 15G 7.2G 68% /
tmpfs 985M 4.0K 985M 1% /dev/shm
none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs 985M 136K 985M 1% /tmp
none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service
tmpfs 197M 124K 197M 1% /run/user/1000

```